

# DOES FINANCIAL DEVELOPMENT HOLD THE KEY TO ECONOMIC GROWTH? THE CASE OF SUB-SAHARAN AFRICA

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## ABSTRACT

The paper investigates the relationship between financial development and economic growth in seven Sub-Saharan Africa countries. Using the panel Granger causality test, the study finds one-way causality running from economic growth to bank developing indicators and a two-way causality between stock market development indicators and economic growth. The fixed-effects estimation shows stock market development has positive and significant effect on economic growth while banking development indicators impact on economic growth is uncertain. Control variables including capital formation, schooling, and life expectancy have positive effect on economic growth. Based on these findings, the study suggests for adoption of policies that create favorable environment for financial market development including efforts to integrate the small capital markets.

**JEL Classification:** G10, O16, O55

**Key words:** economic growth, financial development, stock market development, Sub-Saharan Africa

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## INTRODUCTION

Many factors including internal government policies, political stability, domestic capital formation, development of human capital, banking infrastructure, export policies, foreign direct investment, and foreign aid influence the country's economic growth and provide the most profound platform for economic development. However, for the vast majority of the developing countries especially in sub-Saharan Africa (SSA), these factors are yet to be realized with a majority of their population living in meager subsistence levels (Neelankavil, Stevens, and Roman, 2009). The way forward out of economic stagnation for these countries continues to occupy the work of scholars and other stake holders alike.

The role of financial development<sup>1</sup> in economic development is well discussed in both theoretical and empirical literature. Early in the 20<sup>th</sup> century, scholars like Schumpeter (1934) observed that financial markets play a vital role in the economic growth process by channeling funds to the most efficient investors and by fostering entrepreneurial innovation. Others like Lucas (1988) have dismissed the importance of financial sector's development as a precedent for economic growth noting that economists "badly over-stress" their role.

The development of financial sector especially in developing countries is seen as a part of private sector development strategy to stimulate economic growth and reduce poverty by generating local savings which in turn lead to productive investments and a better allocation of resources. It is supposed to increase investor's borrowing options allowing investors to choose an optimal debt structure for a given period (Mihalca, 2007).

There has been a considerable development of both market and bank-based financial markets in Africa since the early 1990s with substantial financial growth been registered in SSA (Table 1). However, as indicators of stock market development in Table 1 show, SSA markets are still small with few listed companies and low market capitalization. The average number of listed companies on sub-Saharan African markets excluding South Africa is 44 (73 with the inclusion of South Africa) while market capitalization as a percentage of GDP is as low as 0.8% in Uganda. Also, the South Africa's Johannesburg Securities Exchange has about 90% of combined market capitalization of SSA, and with exclusion of South Africa and Zimbabwe the average market capitalization is about 27% of GDP. This is in contrast with other emerging markets like Malaysia with a capitalization ratio of GDP at about 161% (Yartey and Adjasi, 2007).

**TABLE 1. INDICATORS FOR SELECTED STOCK EXCHANGES IN SSA BY 2008**

| Country       | Listed Domestic Companies | Market Capitalization (USD bn) | Value traded (USD bn) | Turnover ratio (%) | Market capitalization (% of GDP) |
|---------------|---------------------------|--------------------------------|-----------------------|--------------------|----------------------------------|
| Côte d'Ivoire | 38                        | 7.1                            | 0.32                  | 4.1                | 30.2                             |
| Ghana         | 35                        | 3.4                            | 0.15                  | 5.2                | 21.1                             |
| Kenya         | 53                        | 10.9                           | 1.44                  | 11.8               | 31.6                             |
| Malawi        | 14                        | 1.8                            | 0.06                  | 3.9                | 41.5                             |
| Mauritius     | 41                        | 3.4                            | 0.4                   | 8.9                | 39.8                             |
| Namibia       | 7                         | 0.6                            | 0.02                  | 2.8                | 7.2                              |
| Nigeria       | 213                       | 49.8                           | 19.95                 | 29.3               | 23.5                             |
| South Africa  | 425                       | 491.3                          | 401.49                | 60.6               | 177.5                            |
| Swaziland     | 7                         | 0.2                            | 0                     | 0                  | 7.8                              |
| Tanzania      | 7                         | 1.3                            | 0.01                  | 2.1                | 6.3                              |
| Uganda        | 6                         | 0.1                            | 0.01                  | 5.2                | 0.8                              |
| Zambia        | 15                        | 2.3                            | 0.07                  | 4.1                | 16.4                             |
| Zimbabwe      | 81                        | 5.3                            | 0.81                  | 5.1                | n.a.                             |

*Source: Standley, 2010*

African stock markets also suffer from low liquidity which is as low as 0.02% in Swaziland. The low liquidity means that it is harder to support a local market with its own trading system, market analysis, brokers, and the like as business is low. However, despite the problems of small size and low liquidity, SSA stock markets have continued to perform remarkably well in terms of return on investment. The Ghana Stock Exchange was rated the world's best-performing market at end of 2004 with a year return of 144% in US dollar terms compared with 30% return by Morgan Stanley Capital International

Global Index (Databank Group, 2004). Within SSA, bourses of Uganda, Kenya, Egypt, Mauritius and Nigeria have been rated as amongst the best annual performers recent years (Yartey and Adjasi, 2007).

Studies have shown that stock market development has a significant positive long-run impact on economic growth through liquidity injection, savings mobilization and equity financing for firms (Demirguc-Kunt and Levine, 1996). Although Sub-Sahara Africa stock market grew by over 300% during the early 2000s, the few studies that have been conducted on financial development and economic growth in Sub-Saharan Africa have not included stock market development as a measure of financial development.

According to Levine and Zervos (1998) study of 47 countries from 1976-93 using cross-country data, stock market liquidity is positively and significantly correlated with current and future rates of economic growth, even after controlling for economic and political factors. They also found that both measures of stock market liquidity and banking development significantly predict future rates of economic growth. They, therefore, concluded that the stock markets provide important but different financial services from those that were offered by banks.

## FINANCIAL DEVELOPMENT IN AFRICA

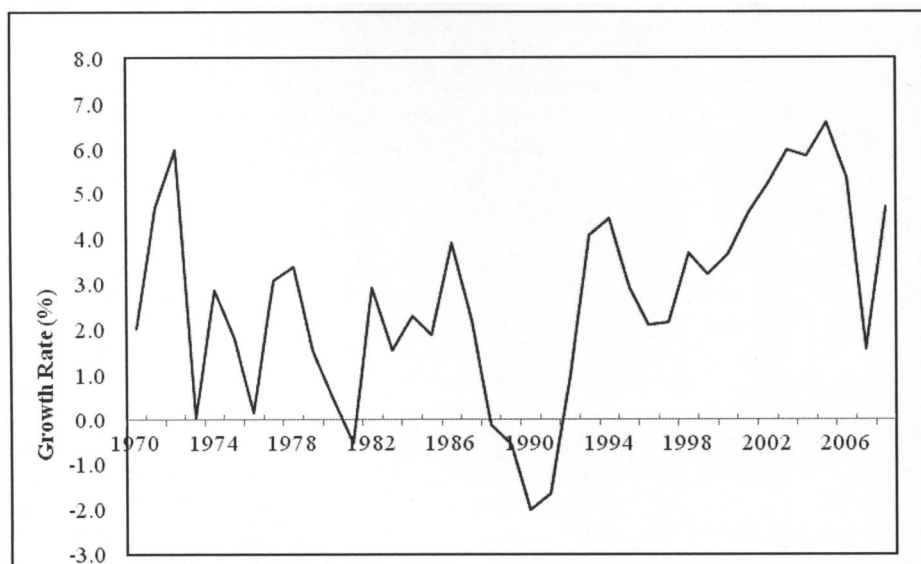
Although there has been a substantial growth in financial services development since the 1990s Africa, SSA countries have faced problems related to the practical functioning of their financial markets. The main challenges are political instability in some economies, high volatility in economic growth, macroeconomic uncertainty, liquidity constraints, underdeveloped trading and settlement structures, limited market information, and lack of adequate supervision by regulatory authorities. Also, since majority of the people in these economies are unable to save or invest due to high levels of poverty the benefits for the overall economy are small and restricted to few and exclusive sectors of the economy (Marone, 2003).

While one of the probable reasons for low inability to save in SSA is the highly volatile economic growth rate which coupled with low economic growth seriously hinder the financial market development as depicted by Figure 1, perhaps the greatest impediment to savings is limited outreach of the financial systems. Banking is very expensive in SSA, as reflected by high spreads between deposits and lending interest rates which act as a disincentive to both savings and lending as it depresses the returns for savers and pushes lending interest rates up. In many SSA countries deposit customers face very high minimum balance requirements and annual fees which partly explain why less than 20 percent of the population in region has a bank account. However, the biggest deterrent to owning a savings account SSA might be the high documentation requirements including the need to present several documents of identification, such as passport, pay slip, utility bill, etc even when large parts of the population live and work in the informal sector (Beck, Fuchs, and Uy, 2009).

Another reason for underdeveloped financial markets in SSA is the small sizes of their economy as increasing evidence suggests that financial market development is related to economy size. Since most of SSA economies are small, efforts to further integrate capital markets and in particular promoting cross-border listings and sharing

of technologies are relevant, in addition to reforms to improve the efficiency of the banking sector and the legal and regulatory environments (Standley, 2010).

**FIGURE 1. SUB SAHARAN AFRICA REAL GDP GROWTH RATE (1970-2008)**



Source: World Bank (2008)

The mal-functioning and ill-developed financial systems are, therefore, impediments not only to mobilization of global capital but also to domestic savings, with adverse consequences on growth and poverty alleviation in Africa (Senbet and Otchere, 2005). However, to sustain the current level of economic growth and encourage both domestic and foreign investment in the continent, Africa needs to rapidly expand, develop and modernize its financial markets. Evidence from recent empirical economic studies suggests that deeper, broader, and better functioning financial markets can stimulate economic growth (Ndikumana, 2001).

Although there are only few studies that have focused explicitly on the link between financial system development and economic growth in Africa, existing evidence suggests that financial development might have a positive effect on economic growth. Odedokun (1996) and Spears (1992) found that aggregate measures of financial intermediation have positive and statistically significant effects on the growth rate of real per capita GDP in Africa. Also, the study by Gelbard and Leite (1999) found positive and statistically significant link between a country's real per capita GDP growth and their indices of financial development. They found that the level of initial financial development and change in the overall financial development index of between 1987 and 1997 were positively and significantly related to economic growth. These results were consistent with studies done elsewhere that suggested that the initial level of financial

development is an important determinant of future economic growth (King and Levine, 1993).

At an empirical level, one of the strongest attempts to investigate the role of financial intermediation in stimulating economic growth in SSA was made by Allen and Ndikumana (2000). Using various indicators of financial intermediation and three different panel techniques they found a positive correlation between financial development and the growth of real per capita GDP for the Southern Africa Development Community (SADC) which includes 15 countries in Central and Southern Africa.

Also, Gries, Kraft, and Meierrieks (2009) study tested for causality between financial deepening, trade openness and economic development for 16 Sub-Saharan African countries using Hsiao-Granger method. Results of their study found only limited support for finance-led growth hypothesis with financial deepening and trade openness having swayed regional development only marginally. They concluded that countries under the study had failed to benefit from financial deepening and recommended a holistic policy approach taking into account other fundamental development factors.

Enisan and Olufisayo (2009) examined the long run and causal relationship between stock market development and economic growth for seven countries in Africa using the autoregressive distributed lag (ARDL) bounds test. Results from their study suggested that stock market development is cointegrated with economic growth in Egypt and South Africa. The Granger causality test within the VECM framework showed unidirectional relations running from stock market development to economic growth. The study suggested that policy makers in the continent should encourage stock market development through appropriate mix of taxes, legal and regulatory policies to remove barriers to stock market operation and thus enhance their efficiency.

All the empirical studies mentioned, except for Enisan and Olufisayo (2009), did not include stock market development in their measure for financial development and by neglecting stock market development variable in their estimates of the link between financial development and economic growth in Africa, the studies cannot be free of the bias associated with omitted variables. Stock market development cannot be ignored as a measure of economic development since it is highly correlated with the development of banks, nonbank financial institutions, insurance companies, and private pension funds. Also, both Allen and Ndikumana (2000) and Enisan and Olufisayo (2009) studies suffer from another weakness in that their models did not include human capital accumulation. This is a huge oversight since growth theory suggests a positive relationship between education and economic growth especially for developing countries (Barro, 1991). This study attempts to correct for this deficiency.

The objective of the study is, therefore, to investigate the relationship between financial market developments and economic growth in selected sub-Saharan African countries for a period of 1991-2007 with the inclusion of stock market development as one of financial development measures. The seven countries were selected based on the availability of data.

The rest of the paper is organized as follows: Section 3 presents model specification and data issues while econometric methodology and empirical findings are presented in Section 4. Concluding remarks are given in the last section.

## **MODEL SPECIFICATION AND DATA DESCRIPTION**

### Model Specification

We assume as in Amavilah (2008) that the value of the  $i^{\text{th}}$  country's economic activity, gross domestic product ( $Y_i$ ), depends on the country's technical capability ( $X_i$ ) such that  $Y_i = F(X_i)$ ,  $i = 1, 2, 3, \dots, N$  countries (1)

Moreover, technical capability has two broad dimensions: domestic ( $X_{i,d}$ ) and global ( $X_{i,f}$ ) links, that is,  $Y_i = F(X_{i,d}, X_{i,f})$ . Domestic production in which global links are, respectively, absent and exogenously present in (2a) and (2b) below:

$$Y_i = A_i F(X_{i,d}) \quad (a)$$

$$Y_i = X_{i,f} F(X_{i,d}) \quad (b) \quad (2)$$

where in (2a)  $A_i \Rightarrow X_{i,f}$ , in (2b)  $X_{i,f} \Rightarrow A_i$ ,  $i$  indicates the  $i^{\text{th}}$  country,  $d$  is for domestic, and  $f$  is the foreign links. In the study, we ignore foreign links since Africa's exporting sectors are largely natural resource based and typically employ very small share of the national labor force. The foreign sector therefore has small or no linkages to the rest of the domestic economy.

The underlying domestic production relations include economically-active population ( $N$ ) and physical capital ( $K$ ) that can be stated in a neoclassical multiplicative form as

$$Y_i = AK_i^\alpha N_i^\beta, \quad \alpha + \beta \leq 1 \quad (3)$$

where  $N$  consists of unskilled labor ( $L$ ) and skilled labor defined as human capital ( $H$ ). Suppose  $N = L^\alpha H^\beta$ , it implies that  $L = (NH^{-\beta})^{1/\alpha}$ . Following Solow's model of economic growth, we assume that  $L$  evolves at an exogenous rate ( $n$ ) equal to the rate of growth of  $N$ , thus

$$L = e^{nt} (NH^{-\beta})^{1/\alpha} \quad (4)$$

Given (4), (3) becomes

$$Y_i = AK^\alpha N^{\beta_1/\alpha} e^{\beta_1 nt} H^{(a\beta_2 - b\beta_1)/\alpha} \quad (5)$$

In per capita and natural logarithmic terms (5) is

$$y_i = a + \alpha k + \beta_1 nt + \left( \frac{a\beta_2 - b\beta_1}{\alpha} \right) h \quad (6)$$

where  $y = \log(Y/N)$ ,  $k = \log(K/N)$ ,  $h = \log(H/N)$ ,  $a$  = productivity shifter,  $\alpha$  = elasticity of physical capital,  $\beta_1$  = elasticity of raw labor, and  $\beta_2$  = elasticity of  $H$ , and  $n$  = natural rate of growth of  $N$  over time  $t$ .

As noted by Curie and Moretti (2003) conventional estimates of the returns to education that focus only on wages understate the total benefits of human capital

accumulation since higher levels of education result in better health for educated individuals and their children. Healthier children go on to be more productive and more educated adults creating inter-generational spillovers which analyses of wage effects alone do not capture. Amavilah (2008) argues that (6) understates the level of  $H$  and overstates its statistical importance. He proposes the correct evolution of  $H$  should be based on the economically active population ( $N$ ), thus

$$H = e^{\Phi q} N \quad (7)$$

where  $q$  is the vector of processes and activities that refine  $N$  such as education, training, experience, health, nutrition, and mortality-fertility. Also,  $n$  is the rate of growth of  $L$  which is equal to the rate of growth of  $N$  based on the exogenous neoclassical growth model.

Given (7), (5) becomes

$$Y_i = AK^a N^{(a\beta_2 + b\beta_1(1-b))/a} e^{(\phi(a\beta_2 - b\beta_1)/a)q + \beta_1 nt} \quad (8)$$

When (8) is normalized we get

$$y_i = a + \alpha k + \beta_1 nt + \left( \frac{\phi(a\beta_2 - b\beta_1)}{a} \right) q \quad (9)$$

and estimate (9) as

$$y_i = a + \alpha k + \beta_1 nt + \eta q + \mu \quad (10)$$

where  $\eta = \frac{\phi(a\beta_2 - b\beta_1)}{a}$

To assess the role of financial development ( $FD$ ) on the economic development of sub-Sahara Africa, this study assumes that financial development enhances growth in  $TFP$ . That is,  $TFP = f(FD)$  since it is widely accepted that  $FD$  is one factor that contributes to higher  $TFP$  growth<sup>2</sup>. Thus, examining the contribution of  $TFP$  to economic growth provides some indication of the influence of financial development in the process of economic development (Ang, 2008).

From (10) we can, therefore, calculate  $TFP$  growth as the “Solow residual” or the portion of output not accounted for by inputs growth.

$$\alpha = y_i - \alpha k - \beta_1 nt - \eta q \quad (11)$$

The  $TFP$  proxy variable ( $\alpha$ ) in (11) is therefore a measure of financial development impact on economic growth. Since  $y$  is per capita GDP, we do not need data for either  $L$  or  $N$ . Instead, we use the rate of population growth ( $n$ ) with the coefficient  $\beta_1 t$ .

## Data Sources and Description

The analysis is based on data for seven sub-Saharan African countries (Botswana, Cote-d'Ivoire, Ghana, Kenya, Nigeria, South Africa, and Swaziland) over the period 1991-2007. The dependent variable, GDP ( $y$ ), is growth in per capita expressed in real U.S. dollars and population ( $N$ ) is expressed in millions of people. Data for  $y$  and  $N$  was obtained online from the World Bank, world development indicators website (World Bank, 2008).

The study includes a set of variables that measure endowments for both physical capital measured as ( $K$ ) and human capital measured as a vector of activities that refine  $N$ . We use average years of schooling ( $SCH$ ) as a measure of  $q$ , the process and activities that refine the active population ( $N$ ). Data for physical capital and average years of schooling was obtained from the World Bank (2008) world development indicators whose data was obtained from Barro and Lee (2000) international data on educational attainment available at the Center for International Development, Harvard University. Since the data covers average years of schooling upto the year 2000, it was extrapolated using the country's real GDP. Data on life expectancy and the natural rate of population growth ( $N$ ) was obtained from the U.S. Census Bureau, International Data Base. Data on gross fixed capital formation ( $K$ ), at constant prices, was obtained from United Nations Statistics Division.

We employ six different measures of financial development to capture the variety of channels through which stock market and banking sector development can affect economic growth. To measure banking sector development, studies traditionally use measures of the overall size of the banking sector by dividing the stock of broad money ( $M2$ ) by GDP. King and Levine (1993) and Levine and Zervos (1998) argue that this indicator does not measure whether the liabilities are those of banks, the central bank, or other financial intermediaries. Thus, we follow King and Levine (1993) and Boyd et al. (2001) and employ three proxies to measure banking sector development. The first measure is the ratio of liquid liabilities ( $LIQ$ ) of the financial sector to GDP. This ratio equals the sum of currency, demand and interest-bearing liabilities of bank and non-bank financial intermediaries as a percentage of GDP. This indicator measures the size of the banking sector in relation to the economy as a whole and has been found to be strongly associated to the real GDP per capita (King and Levine, 1993).

As noted by King and Levine (1993),  $LIQ$  measure may not be closely related to other financial services such as risk management and information processing. Consequently, our second measure is bank assets ( $BA$ ), which equals the ratio of total assets of deposit money banks (commercial banks and other deposit taking banks) as a percentage of GDP. Boyd et al. (2001) point out that this ratio reflects the overall level of development of the banking sector as it measures the importance of deposit money banks, as reflected in their total assets relative to the economy. Since the above two measures do not consider the allocation of capital between the private and public sector (King and Levine, 1993), and in order to obtain a more direct measure of financial depth, private credit ( $PC$ ) is our third measure of banking sector development. This measure equals the value of loans made by banking institutions to the private sector as a percentage of GDP. The primary advantage of this measure is that, by isolating credit to the private sector, it measures more precisely the contribution of financial institutions in funding private sector investment (Levine and Zervos, 1998; Boyd et al., 2001; Beck and Levine, 2003). Stock market development measures are also included in the model specification. In

order to capture the relative size and the level of development of the stock market, the first measure is total stock market capitalization as a percentage of GDP (*SMC*). This measure is computed as the average value of listed domestic shares on domestic exchanges in a year as a share of the size of the economy. Studies frequently use *SMC* as an indicator of stock market development (Levine and Zervos, 1998; Boyd et al., 2001). The second and third measures of stock market development are used in the literature as indicators of market liquidity. Value traded (*VT*) equals the value of the trades of domestic shares on domestic exchanges divided by GDP. *VT* measures trading volume as a share of national output and should therefore positively reflect liquidity on an economy wide basis (Levine and Zervos, 1998). As noted by Boyd et al. (2001), *VT* complements *SMC* because it reflects the actual volume of market transactions along with the overall size of the market. Turnover ratio (*TR*) equals the value of the trades of domestic shares on domestic exchanges divided by the value of listed domestic shares. Thus, *TR* measures trading volume relative to the size of the market. This ratio complements the *SMC*. A large but inactive market will have a large market capitalization ratio, but a small turnover ratio (Boyd et al., 2001). The dataset used for financial development indicators are obtained from Financial Structure and Economic Development Database (FSEDD). Table 2 provides a more detailed description of all variables used and their data sources while Table 3 presents descriptive statistics for all variables used in the analysis.

TABLE 2. DEFINITION OF VARIABLES

| Variable                          | Definition                                                                   | Source |
|-----------------------------------|------------------------------------------------------------------------------|--------|
| Economic Growth (Y)               | Growth in Per capita real GDP                                                | WDI    |
| Stock Market Capitalization (SMC) | Total market value of all listed shares / GDP                                | FSEDD  |
| Stock Market Turnover Ratio (TR)  | Value of trades of shares over market capitalization                         | FSEDD  |
| Stock Market Value Traded (VT)    | Value of trades of domestic stocks / GDP                                     | FSEDD  |
| Private Credit (PC)               | Private Credit by Deposit Money Banks and Other Financial Institutions / GDP | FSEDD  |
| Liquid Liabilities (LIQ)          | Liquid Liabilities / GDP                                                     | FSEDD  |
| Bank Assets (BA)                  | Deposit Money Bank Assets / GDP                                              | FSEDD  |
| School Enrollment (SCH)           | Average years of Schooling                                                   | B&L    |
| Capital Formation (K)             | Net additions of capital stock that are used to provide goods and services.  | UN     |
| Natural growth (N)                | The births and deaths in a country's population                              | WDI    |
| Life Expectancy (L)               | The average number of years a newborn infant would be expected to live.      | WDI    |

TABLE 3. DESCRIPTIVE STATISTICS

|      | <i>Y</i> | <i>SMC</i> | <i>VT</i> | <i>TR</i> | <i>BA</i> | <i>LIQ</i> | <i>PC</i> | <i>SCH</i> | <i>K</i> | <i>N</i> | <i>L</i> |
|------|----------|------------|-----------|-----------|-----------|------------|-----------|------------|----------|----------|----------|
| Mean | 3.08     | 0.33       | 0.11      | 0.09      | 0.26      | 0.29       | 0.30      | 5.4        | 0.20     | 2.05     | 53.32    |
| Std. |          |            |           |           |           |            |           |            |          |          |          |
| Dev. | 0.39     | 0.49       | 0.33      | 0.13      | 0.17      | 0.10       | 0.35      | 1.6        | 0.85     | 0.77     | 7.67     |
| Obs. | 119      | 119        | 119       | 119       | 119       | 119        | 119       | 119        | 119      | 119      | 119      |

## EMPIRICAL RESULTS AND DISCUSSION

The study uses panel regression analysis for the broader determinants of economic growth. While panel regression analysis can give the effect through the degree of association between financial development and economic growth, it cannot depict the direction of causality and long run relationship between financial development and economic growth. Thus, in order to determine the direction of causality between financial development and economic growth the study first performs Granger causality tests.

### Causality Analysis

Before proceeding to the identification of a possible long run relationship we need to verify that all variables are integrated of order one in levels. We use panel unit root tests as in Im, Pesaran, Shin (2003) denoted as IPS. In the test, the null hypothesis is that of a unit root where the IPS statistic is based on averaging individual Dickey-Fuller unit root tests ( $t_i$ ) according to

$$t_{IPS} = \frac{\sqrt{N}(\bar{t} - E[t_i | \rho_i = 0])}{\sqrt{\text{var}[t_i = 0]}} \rightarrow N(0,1) \quad (12)$$

where  $\bar{t} = N^{-1} \sum_{i=1}^N t_i$ . The moments of  $E[t_i | \rho_i = 0]$  are obtained by Monte Carlo simulation and tabulated in IPS.

To test the causality between economic growth and financial development measures our methodology follows Hurlin and Venet (2001) and considers two covariance stationary variables, denoted by  $x$  and  $y$ , observed on  $T$  periods and on  $N$  cross-section units. Given the standard Granger (1969) causality definition, for each individual  $i \in [1, N]$ , we say that the variable  $x_{i,t}$  is causing  $y_{i,t}$  if we are better able to predict  $y_{i,t}$  using all available information than if the information apart from  $x_{i,t}$  had been used. In practice, it will not usually be possible to use completely optimum predictors, so we consider only linear ones. For this reason, we consider a time-stationary VAR representation, adapted to a panel data context. For each individual  $i$  we have,  $\forall t \in [1, T]$ :

$$y_{i,t} = \sum_{k=1}^p \gamma^{(k)} y_{i,t-k} + \sum_{k=0}^p \beta_i^{(k)} x_{i,t-k} + \mu_{i,t} \quad (13)$$

where  $v_{i,t} = \alpha_i + \varepsilon_{i,t}$  and  $\varepsilon_{i,t}$  are *i.i.d*  $(0, \sigma_\varepsilon^2)$ . We assume that the autoregressive coefficients  $\gamma^{(k)}$  and the regression coefficients slopes  $\beta_i^{(k)}$  are constant  $\forall k \in [1, p]$ . It is also assumed that parameters  $\gamma^{(k)}$  are identical for all individuals while regression coefficients slopes  $\beta_i^{(k)}$  could have an individual dimension. Thus, we estimate a panel data model with fixed coefficients whose residuals are assumed to satisfy the standard properties.

As noted by Hurlin and Venet (2001), the use of panel data dimension has a number of advantages including providing a large number of observations thereby increasing the degrees of freedom. It also reduces collinearity among explanatory variables which combined with the above improves the efficiency of Granger causality tests.

The study assumes balanced panels and lag orders ( $K$ ) identical for all cross-units while respecting the  $T > 5 + 2K$  condition which guarantees the validity of the proposed test with short  $T$  samples (Hurlin and Venet, 2001). The study uses  $F$ -tests to test for Granger non-causality and begins by testing the following hypothesis:

$$H_0 = \beta_i^{(k)} = 0 \quad \forall i \in [1, N], \forall k \in [1, p] \quad (14a)$$

$$H_1 \neq \beta_i^{(k)} = 0 \quad \forall i \in [1, N], \forall k \in [0, p] \quad (14b)$$

The test statistic for the panel Granger Causality Tests in (14) are computed following the equation:

$$F = \frac{(SSR_r - SSR_u)/p}{SSR_u/[NT - N(1 + 2p) + p]} \quad (15)$$

where  $SSR_r$  is the sum of squared residues found in (14) when the nullity of the  $k$  coefficients associated with the variable  $x_{i,t-k}$  only for the cross-section unit  $i$  are imposed and  $NT$  is the total number of observations. These  $N$  individual tests identify the cross-section unit for which there are no causality relationships. If the hypothesis in (14) is not rejected, this means that there exists a subgroup of cross-section units for which the variable  $x$  does not cause the variable  $y$ . This hypothesis can be analyzed as the consequence of the heterogeneity of the data-generating process. The causality relationship is relevant only for a subgroup of cross-section units.

### Panel Unit Root Results

The variables of economic growth ( $y$ ), measures of financial development ( $LIQ$ ,  $BA$ ,  $PC$ ,  $SMC$ ,  $VT$ ,  $TR$ ), and all control variables ( $K$ ,  $N$ ,  $SCH$ ,  $L$ ) are tested for unit root both in levels and in first differences. The results from the Im, Pesaran, Shin (2003) panel unit root (IPS) test are presented in Table 4 and are reported with a trend<sup>3</sup>. The results from the IPS test indicate that  $N$  and  $L$  are statistically significant at their level, implying that they are integrated of order I (0). As can be inferred from this table, we cannot reject the unit root hypothesis for  $y$ ,  $LIQ$ ,  $BA$ ,  $PC$ ,  $SMC$ ,  $VT$ ,  $TR$ ,  $K$ , and  $SCH$  when they are taken in levels, but when first differences are used, the hypothesis of unit root is rejected at the 1%, 5%, and 10% significance levels. Therefore, these variables are well characterized as I (1) process. These results allow us to proceed on with the panel data estimation method and the Granger causality tests between economic growth and financial development variables in consideration.

### Panel Causality Results

In this section, we examine the causal relationships between economic growth and financial development measures. In particular, does a well-developed financial sector lead to higher economic growth as the supply-leading view hypothesizes? Alternatively, does financial development follow economic growth as the demand-pulling view hypothesizes? First, we investigate the causal interaction between banking sector development indicators and economic growth. The empirical results from the Granger causality between banking sector development indicators and economic growth are presented in Table 5. The Granger causality results show that the null hypothesis that economic growth does not cause financial development is rejected for all banking sector

development indicators. Therefore, there is an evidence of one-way causality running from economic growth to *BA*, *LIQ*, and *PC*. Furthermore, there is no evidence that *BA* and *LIQ* have an effect on economic growth. However, there is short-run Granger causality running from *PC* to economic growth at the 10% level. Thus, there is a bi-directional relationship between *PC* and economic growth, which is consistent with earlier studies suggesting that bank credit to the private sector stimulates economic growth through improving efficiency (King and Levine, 1993).

TABLE 4. IPS PANEL UNIT ROOT TESTS

| Variable | Levels    | First difference |
|----------|-----------|------------------|
| Y        | 0.624     | -2.840***        |
| SMC      | -0.571    | -1.635*          |
| TR       | -1.142    | -4.441***        |
| VT       | 2.588     | -1.850**         |
| PC       | 0.216     | -1.206**         |
| LIQ      | 0.864     | -1.604*          |
| BA       | -1.243    | -3.334***        |
| K        | 0.325     | -1.862**         |
| N        | -3.445*** | 0.221            |
| SCH      | 3.382     | -1.466*          |
| L        | -4.331*** | -4.076***        |

*Notes: Panel unit root tests are estimated based on Im, Peseran and Shin (2003) test. \*\*\* signifies rejection of the unit root hypothesis at the 1% level.*

TABLE 5. PANEL GRANGER CAUSALITY TESTS (BANKING DEVELOPMENT INDICATORS)

| Null Hypothesis              | F-statistics | p-value   |
|------------------------------|--------------|-----------|
| BA does not Granger cause Y  | 1.1969       | 0.3140    |
| Y does not Granger cause BA  | 3.09302      | 0.06462*  |
| LIQ does not Granger cause Y | 2.45188      | 0.10833   |
| Y does not Granger cause LIQ | 4.54822      | 0.02166** |
| PC does not Granger cause Y  | 2.069        | 0.0869*   |
| Y does not Granger cause PC  | 4.2438       | 0.0027*** |

*Notes: Dynamics asterisks indicate rejection of the null hypotheses that there is no causal relationship between the two variables. \*\*\*Significant at the 1% level.*

The evidence of one-way causality running from economic growth to financial development, as measured by banking sector development indicators, gives support to the idea that financial development may be a result of economic growth or demand-pulling phenomenon. According to Patrick (1966), strong economic growth leads to higher demand for various types of financial services. Hence, more financial institutions,

products, and services emerge in the markets to meet the increased demand. Overall, the results of the Granger causality tests suggest that the banking sector is not developed in the region sufficiently to support sustained economic development. Therefore, the conclusion of King and Levine (1993) that “finance seems importantly to lead economic growth” (p.730), is not supported by our research findings. This is not to say that banking sector development does not promote economic growth, because both banking development and economic growth appear to have two-way causality when PC is used as a measure of banking sector development.

Table 6 summarizes the Granger causality results between stock market development indicators and economic growth. The results show bi-directional relationships between the three indicators and economic growth. As evident from Table 6, the null hypotheses that *SMC*, *VT*, and *TR* do not cause economic growth are rejected. The results also show the presence of causality from economic growth to stock market development. These results reveal strong bi-directional causality relationships between stock market development and economic growth in sub-Saharan Africa.

The bi-directional causality relationships between stock market development and economic growth suggest that economic growth initially stimulates and permits financial development and, in turn, financial development results in accelerated economic growth. In general, our findings provide some support for the argument that financial development will originally grow in response to services that is directed to the financial systems by the economic growth. Once this stage is reached, the direction of causality is reversed and financial development directly stimulates economic growth. That is, the evolution of the financial sector (specifically the stock market) tends to be more likely to stimulate and promote economic growth.

**TABLE 6. PANEL GRANGER CAUSALITY TESTS  
(STOCK MARKET INDICATORS)**

| Null Hypothesis              | <i>F</i> - Statistics | <i>p</i> -value |
|------------------------------|-----------------------|-----------------|
| SMC does not Granger cause Y | 3.4157                | 0.0020**        |
| Y does not Granger cause SMC | 2.4463                | 0.0678*         |
| VT does not Granger cause Y  | 4.2603                | 0.0064***       |
| Y does not Granger cause VT  | 2.4966                | 0.0077***       |
| TR does not Granger cause Y  | 4.8712                | 0.0032***       |
| Y does not Granger cause TR  | 3.4157                | 0.0026**        |

*Notes: Dynamics asterisks indicate rejection of the null hypotheses that there is no causal relationship between the two variables. \*\*\*Significant at the 1% level.*

**Panel Regression Results**

One of the challenges with the use of cross-sectional data in regression estimation is that it does not allow for control of country-specific fixed effects and could lead to potential biases (Islam, 1995). Thus, in order to explore the relationship between financial development and economic growth we use annual data to estimate the contemporaneous non-dynamic fixed effects panel model of economic growth by regressing economic growth control variables (*X*) and financial development indicators in the following format

$$Growth = \beta_0 + \alpha X + \Phi \text{ stocks market indicators} + \varphi \text{ Banks Indicators} + \mu \quad (16)$$

After combining (11) and (17), the following panel regression model is estimated

$$y_{it} = \beta_0 + \beta_1 K_{it} + \beta_2 N_{it} + \beta_3 SCH_{it} + \beta_4 L_{it} + \beta_5 SMC_{it} + \beta_6 VT_{it} + \beta_7 TR_{it} + \beta_8 BA_{it} + \beta_9 PC_{it} + \beta_{10} LIQ_{it} + \mu \quad (17)$$

where all coefficients are expected to be positive. All variables are estimated in natural log except  $N$  and  $SCH$  which are already in percentages.

Financial development measures include banks indicators (Bank Assets, Private Credit, and Liquid Liabilities) and Stock Market Indicators (Stock Market Capitalization, Stock Market Turnover Ratio, and Stock Market Value Traded). Model 1 estimates growth as a function of the control variables as described in (11). Model 1 plus banks financial indicators equals model 2, while model 1 plus stock market financial indicators equals model 3. Model 4 estimates the control variables and financial development indicators. The results of all the regressions are listed on Table 7. Since model 4 includes all the variables with the highest adjusted  $R^2$  and all the variables except banking assets did not change much across the models, it is the model of discussion in the study.

As summarized in table 7, all panel regression models have  $R^2$  that range from 0.64 to 0.82, an indication of good predictive power. The control variables have the predicted signs and are significant at 1% level in all the three models. Model 4 which includes all the indicators will be the model for discussion. The results of model 4 indicate that capital formation has a positive impact on economic growth in Sub-Saharan Africa. Specifically, the results indicate that a 10% increase in gross capital formation, *ceteris paribus*, will results in 1.6% in GDP growth in Sub-Saharan Africa. The results also indicate that average years of schooling and increases in life expectancy all have positive impact on economic growth in Sub-Saharan Africa. However, the results indicate that increases in population growth will lead to lower GDP growth. These results are consistent with United Nations Population Fund (UNPFA) 2009 report which notes that high population growth rates and poverty are strongly correlated. The report notes that lower population growth in developing countries may reduce the pressure on national resources and the need for social investments, making it possible for countries to invest in human and physical capital that will result in enhanced economic growth (UNPFA, 2009).

The effects of financial development on economic growth, as denoted by the signs of their coefficients, were mixed. The results indicate that stock market capitalization ( $SMC$ ) and private credit ( $PC$ ) growth have positive impact on economic growth. Specifically, a 10% growth in  $SMC$  and  $PC$ , *ceteris paribus*, will lead to 2.4% and 5.9% GDP growth, respectively. Bank Assets ( $BA$ ) growth, although having a significant coefficient, have a negative coefficient. Also, stock market turnover ratio ( $TR$ ), value traded ( $VT$ ), and banking sector liquid liabilities ( $LQ$ ) have no effect on economic growth.

The significant  $SMC$  (capitalization) and  $PC$  (private capital) coefficients accompanied by other non-significant coefficients might be a reflection of lack of broad and deep financial markets in SSA. As noted by Levine and Zervos (1998)  $SMC$  does not necessarily foster resource allocation. Rather, it is the ability to trade the economy's productive technologies easily that influences resource allocation and growth. Also, with only 1 in 5 households having access to finance there are low levels of  $PC$  access in SSA

which is just 18% of GDP. This is occasioned by weak propensity to lend by the banks which has been blamed on limitations to secure lending, weak enforcement of creditor rights, and inadequate debtor information systems (Sacerdoti, 2005).

The finding of a weak relationship between  $TR$ ,  $VT$ , and  $LQ$  financial development variables and economic growth in Sub-Saharan Africa is consistent with earlier studies from the same sub-region. Allen and Ndikumana (2000) study on financial intermediation and economic growth among the Southern Africa Development Community (SADC) countries, which are part of Sub-Saharan Africa countries, argue that the weak link between financial intermediation and economic growth might reflect pervasive inefficiencies in the credit allocation mechanism among these countries. They recommend the strengthening of financial sector legislation and banking system supervision as one way of enhancing the financial sector efficiency. A study by Aziakpono (2005) that examined the effects of financial integration on financial development and economic performance of the Southern African Customs Union (SACU) countries also found a weak relationship between financial intermediation and economic growth. He attributed this to, among others, weak institutional and structural impediments within the countries.

**TABLE 7. PANEL REGRESSION RESULTS**

**Dependent variable:  $y_{it}$**

**Models**

| <b>Variable</b>     | <b>1</b>             | <b>2</b>             | <b>3</b>             | <b>4</b>             |
|---------------------|----------------------|----------------------|----------------------|----------------------|
| Intercept           | 2.90<br>(75.09)***   | 2.91<br>(76.5)***    | 2.86<br>(73.15)***   | 3.23<br>(68.04)***   |
| K                   | 0.063<br>(4.69)***   | 0.057<br>(4.22)***   | 0.061<br>(4.69)***   | 0.163<br>(7.49)***   |
| N                   | 0.251<br>(2.88)***   | 0.261<br>(2.92)***   | 0.029<br>(3.61)***   | -0.261<br>(-6.99)*** |
| SCH                 | 0.045<br>(13.06)***  | 0.043<br>(11.43)***  | 0.052<br>(14.21)***  | 0.143<br>(12.95)***  |
| L                   | -0.002<br>(-2.81)*** | -0.003<br>(-3.06)*** | -0.002<br>(-2.84)*** | 0.045<br>(17.72)***  |
| SMC                 |                      |                      | 0.046<br>3.28)***    | 0.242<br>(1.97)**    |
| VT                  |                      |                      | -0.019<br>(-1.18)    | -0.198<br>(-1.41)    |
| TR                  |                      |                      | 0.022<br>(0.59)      | 0.368<br>(1.01)      |
| BA                  |                      | 0.091<br>(2.094)**   |                      | -0.781<br>(-2.53)*** |
| PC                  |                      | -0.097<br>(-2.51)**  |                      | 0.598<br>(2.539)**   |
| LIQ                 |                      | 0.061<br>(1.27)      |                      | 0.281<br>(1.023)     |
| Adj. R <sup>2</sup> | 0.64                 | 0.77                 | 0.75                 | 0.82                 |

*t-values in parenthesis, \*\*\* Significant at 1%*

## SUMMARY AND CONCLUSIONS

The paper empirically investigates the relationship between financial development and economic growth using cross-country panel data analysis of 7 Sub-Saharan Africa countries. We look at the aggregate indicators of economic growth; and separately at the financial development indicators that include bank based and stock market based indicators. Like many other studies in this field, the empirical findings suggest that capital formation, increased level of schooling, and increase in life expectancy play an important role in enhancing economic growth in Sub-Saharan Africa. However, increases in population might translate to the lowering on economic growth in the same region.

The granger causality results suggest that financial development may be a result of economic growth. Our findings provide some support for the argument that financial development will originally grow in response to services that are directed to the financial systems by the economic growth. Once this stage is reached, the direction of causality is reversed and financial development directly stimulates economic growth. Granger causality tests suggest that the financial sector may not be sufficiently developed in the region to support sustained economic development.

The results of the study also indicate that stock market development and banking sector development, especially in private credit sector, will lead to enhanced economic growth although the development of other financial intermediaries do not seem to explain growth in the Sub-Saharan Africa. The study points out a mixed relationship between financial development and economic growth in Sub-Saharan Africa. One of the explanations for this finding could be that low economic growth leads to low levels of savings which might have led to the weakness in the financial development. Another reason might be the lack of reliable data, which by itself might affect the viability and reliability of our results.

Based on these findings, the study suggests for development of strong, broad, and deep financial markets so that they can reach a larger segment of the population in their economies. In particular, SSA countries should adopt financial sector policies that achieve efficient markets. In addition, efforts to integrate the small capital markets especially promotion of cross-border listings and sharing of technologies should be encouraged to overcome low market liquidity. Also, investment in human capital- especially in education and health- should be a top priority as this will enhance the utilization (productivity) of both physical and financial capital which is known to foster economic growth.

## ENDNOTES

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<sup>1</sup> Financial development includes the expansion of financial services and the growth of financial institutions as well as an increase in per capital amount of financial services and institutions or an increase in the ratio of financial assets to income (Ahmed and Ansari, 1998).

<sup>2</sup> It has been noted that efficient financial system performs the task of screening investment projects which improves the quality of investments, reduces business failures and effectively raises productivity (Greenwood and Jovanovic, 1990).

<sup>3</sup> The lag length used to compute the IPS panel unit root test statistics is based on Akaike Information Criteria (AIC).

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